**DE LA SALLE UNIVERSITY-DASMARINAS**
**COLLEGE OF INFORMATION AND COMPUTING SCIENCE**
**COMPUTER SCIENCE DEPARTMENT**

# COURSE SYLLABUS

**Second Semester, SY 2024-2025**

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| **COURSE CODE** | **:** | S-ITCS227 |
| **COURSE TITLE** | **:** | APPLICATION DEVELOPMENT AND EMERGING TECHNOLOGIES |
| **COURSE TYPE** | **:** | Lecture |
| **COURSE CREDIT** | **:** | Two units |
| **MODALITY** | **:** | Traditional Blended Learning |
| **PRE-REQUISITES** | **:** | None |
| **CO-REQUISITES** | **:** | None |
| **PROFESSOR** | **:** | Ms. Erika Kyle D. Colocado |
| **CONSULTATION** | **:** | M/1000-1100/ COS-100C/ ONSITE<br>W/ 1300-1500/ COS-100C/ ONSITE |
| **CONTACT DETAILS** | **:** | edcolocado@dlsud.edu.ph (MS Teams) |

## VISION:

A leading Catholic university that inspires excellence and drives innovation towards a just, peaceful, and sustainable society.

## MISSION:

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension program.

## COURSE DESCRIPTION:

This course focuses on the development of applications using web, mobile and emerging technologies with emphasis on requirements management, interface, design, usability, testing and deployment, including ethical and legal considerations.

**COURSE LEARNING OUTCOMES:**

By the end of this course, students can:

CLO1. interpret the difference of front-end and back-development in web application;
CLO2. understand Web Development concept by dealing client-based case study;
CLO3. implement Object Oriented way of programming;
CLO4. manage website using Model, View and Control;
CLO5. develop and deploy complete Web Application in live server for pilot testing; and
CLO6. demonstrate 21st century skills in all learning activities.

**LEARNING PLAN:**

**Learning Outcomes for Midterm Period**

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| CLO1 | Interpret the difference of front end and back-development in web application. | TLO1 | Learn and differentiate different web development software and its features |
| CLO2 | Understand Web Development concept by dealing client-based case study. | TLO2 | Obtain tools in developing Web Application Project. Students can implement objects in their codes. |
| CLO3 | Implement Object Oriented way of programming. | TLO3 | Design web application both front-end and back-end. Students can know and use different OOP programming techniques. |
| CLO4 | Manage website using Model, View and Control. | TLO4<br><br>TLO5 | Design and Implements server-side controls.<br><br>Design and Implement Master page. |
| CLO5 | Develop and deploy complete Web Application in live server for pilot testing. | TLO6 | Create and Design Web application report tools. |
| CLO6 | Demonstrate 21st century skills in all learning activities. | TLO7 | Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real- World Problem- Solving, Use of ICT |

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| | | | (not part of topic, but integrated in assessments) |

| Module No. and Learning Outcomes | Week/Date | Teaching-Learning Activities/Assessment Strategies | Technology Enabler | Onsite | Asynchronous | Allotted Hours |
|---|---|---|---|---|---|---|
| Module 0 | Week 1:<br>Jan. 20 - 25 | • Gospel Reading, Reflection and Prayer<br>• **Course Orientation**<br>  ○ Course Syllabus<br>  ○ Course Overview/Orientation<br>• **Class Attendance/Participation** | Schoolbook<br>PowerPoint | ✓ | | 2 |
| Module 1<br><br>CLO1<br>CLO2<br>CLO4<br>CLO5<br>CLO6<br><br>TLO1<br>TLO2<br>TLO7 | Week 2:<br>Jan. 27 – Feb. 1 | • Gospel Reading, Reflection and Prayer<br>• **Module 1: Introduction to APIs**<br>  ○ Understanding APIs and their role in modern web applications<br>• **Enabling Assessment: Seatwork #1**<br>  ○ A 10-points Seatwork (essay) about Module 1: Introduction to APIs | Schoolbook<br>PowerPoint | ✓ | | 2 |
| Module 2 | | • Gospel Reading, Reflection and Prayer | | | ✓ | 2 |

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| CLO1<br>CLO2<br>CLO4<br>CLO5<br>CLO6<br><br>TLO1<br>TLO2<br>TLO6<br>TLO7 | Week 3:<br>Feb. 3 – Feb. 8 | • **Module 2: Understanding APIs and web development**<br>  ○ Introduction to client-side development (HTML, CSS, JavaScript)<br>  ○ Understanding the client and server using APIs<br>  ○ Basic concepts of RESTful APIs<br>  ○ Creating RESTful APIs with CRUD operation (Create, Read, Update, Delete)<br><br>• **Enabling Assessment: Seatwork#2**<br>  ○ A 10-items Seatwork (multiple choice) about Module 2: Understanding APIs and Web development | Schoolbook<br>PowerPoint | | | |
| | Week 4:<br>Feb. 10 – Feb.15 | • Gospel Reading, Reflection and Prayer<br><br>• **Enabling Assessment: Quiz**<br>  ○ A 10-items Seatwork (Fill-in the blanks about Module 2: Understanding APIs and Web development | Schoolbook<br>PowerPoint | | ✓ | 2 |
| Module 3<br><br>CLO3<br>CLO6 | Week 5:<br>Feb. 17 – Feb. 22 | • Gospel Reading, Reflection and Prayer<br><br>• **Module 3 (Part 1): Backend Development** | Schoolbook<br>PowerPoint | ✓ | | 2 |

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| TLO1<br>TLO3<br>TLO6<br>TLO7 | | ○ Object-oriented programming concepts applied to server-side development<br>○ Understanding backend frameworks | | | | |
| | Week 6:<br>Feb. 24 – Mar. 1 | • **Enabling Assessment: A 40-items Long quiz about:**<br>○ Module 1: Introduction to APIs<br>○ Module 2: Understanding APIs and web development<br>○ Module 3 (Part 1): Backend Development | Schoolbook | | ✓ | 2 |
| Module 3<br><br>CLO3<br>CLO6<br><br>TLO1<br>TLO3<br>TLO6<br>TLO7 | Week 7:<br>Mar. 3 – Mar. 8 | • Gospel Reading, Reflection and Prayer<br>• **Module 3 (Part 2): Backend Development**<br>○ Object-oriented programming concepts applied to server-side development<br>○ Understanding backend frameworks | Schoolbook<br>PowerPoint | ✓ | | 2 |
| Module 4<br><br>CLO2<br>CLO4<br>CLO6<br><br>TLO1 | Week 8:<br>Mar. 10 – Mar. 15 | • Gospel Reading, Reflection and Prayer<br>• **Module 4: Master pages**<br>• **Class Participation:**<br>○ A 10-items Short quiz about Module 5: Master pages | Schoolbook<br>PowerPoint | | ✓ | 2 |

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| TLO2<br>TLO4<br>TLO5<br>TLO6<br>TLO7 | | | | | | |
| | Week 9:<br>Mar. 17 – Mar. 22 | **MIDTERM SUMMATIVE ASSESSMENT**<br>**Coverage:**<br>○ Module 1: Introduction to APIs<br>○ Module 2: Understanding APIs and web development<br>○ Module 3: Backend Development | Pen and Paper | ✓ | | 2 |
| | | | **TOTAL:** | 10 | 8 | 18 |

**Learning outcomes of Final Period:**

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| CLO5 | Create a Python application that accesses an API based on user input and processes and displays the JSON data that is returned. | TLO5 | Understand and learn APIs and JSON. |
| CLO7<br><br><br><br><br>CLO8 | Evaluate the importance of working with your instructor and fellow students to make the material in this course applicable to the local situation.<br><br>Demonstrate 21st-century skills in all learning activities. | TLO7 | Application of the course material to local situations. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of the topic, but integrated into assessments) |

| Module No. and Learning Outcomes | Week/Date | Teaching-Learning Activities/Assessment Strategies | Technology Enabler | Onsite | Asynchronous | Allotted Hours |
|---|---|---|---|---|---|---|
| Module 5<br><br>CLO5<br>CLO7<br>CLO8<br><br>TLO5<br>TLO7 | Week 10:<br>Mar. 24 – Mar. 29 | • Gospel Reading, Reflection and Prayer<br><br>• **Module 5: Introduction to Database**<br>o Databases concepts and connectivity<br><br>• **Enabling Assessment: Short quiz #1**<br>o A 10-items Short quiz (Fill-in the blanks about Module 6: Introduce to Database<br><br>• **Sustainability Week (Online class)** | Schoolbook<br>PowerPoint | | | 2 |
| Module 6<br><br>CLO5<br>CLO7<br>CLO8<br><br>TLO5<br>TLO7 | Week 11:<br>Mar. 31 – Apr. 5 | • Gospel Reading, Reflection and Prayer<br><br>• **Module 6: Understanding the real-time applications**<br><br>• **Enabling Assessment: Seatwork #1**<br>o A 10-items Seatwork (Multiple choice) about Module 7: Understanding the real-time applications | Schoolbook<br>PowerPoint | ✓ | | 2 |

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| Module 7<br><br>CLO5<br>CLO7<br>CLO8<br><br>TLO5<br>TLO7 | Week 12:<br>Apr. 7 – Apr. 12 | • Gospel Reading, Reflection and Prayer<br><br>• **Module 7 (Part 1): Emerging technologies**<br>  ○ Understand how technologies emerging<br>  ○ Emerging technologies concepts and features<br>• **Enabling Assessment: Assignment #1**<br>  ○ A 15-items Assignment (True or False (10 points) and Essay (5 points) about Module 8: Emerging technologies | Schoolbook<br>PowerPoint | | ✓ | 2 |
| | Week 13:<br>Apr. 14 – Apr. 19 | Independent Learning Days<br>(Apr. 14 – Apr. 15)<br><br>Holy Week (Apr. 16 – Apr. 19) | | | ✓ | 2 |
| | Week 14:<br>Apr. 21 – Apr. 26 | Research week | | | ✓ | 2 |
| | Week 15: | • **Enabling Assessment: A 40-items Long quiz about:**<br>  ○ Module 4: Master pages<br>  ○ Module 5: Introduction to databases | Pen and<br>Paper | ✓ | | 2 |

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| | Apr. 28 – May 3 | ○ Module 6: Real-time applications | | | | |
| Module 7<br><br>CLO5<br>CLO7<br>CLO8<br><br>TLO5<br>TLO7 | Week 16:<br>May 5 – May 10 | • Gospel Reading, Reflection and Prayer<br><br>• **Module 7 (Part 2): Emerging technologies**<br>○ Emerging technologies concepts and features<br><br>• **Enabling Assessment: Seatwork #2**<br>○ A 10-items (Identification) about Module 8 part two: Emerging technologies | | ✓ | | 2 |
| | Week 17:<br>May 12 - 17 | • Review week in preparation for the Final Examination<br><br>• Completion of assessments/requirements | | | ✓ | 2 |
| | Week 18:<br>May 13 – May 24 | **FINALS SUMMATIVE ASSESSMENT**<br>Coverage:<br>○ Module 5: Introduction to databases<br>○ Module 6: Real-time applications<br>○ Module 7: Emerging technologies | Pen and Paper | ✓ | | 2 |
| | | | **TOTAL:** | 8 | 10 | 18 |

**GRADING SYSTEM:**

**Midterm**

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|---|---|
| Enabling Assessments and Community Service Participation | - 50% |
| Attendance/Class Participation | - 20% |
| Summative Assessments | - 30% |
| | **100%** |

**Finals**

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|---|---|
| Enabling Assessments and Community Service Participation | - 50% |
| Attendance/Class Participation | - 20% |
| Summative Assessments | - 30% |
| | **100%** |

**Overall Final Grade = Midterm + Final/2**

## COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy

a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
b. The onsite meeting will only be held at the schedule indicated in the registration form.
c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

**REFERENCES:**

Online References:


| Call number/ E-provider | Reference Material |
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| | insights.com/innovators-guide/emerging-technologies-full-guide/ |

**Prepared: Computer Science Department**

**2nd Semester 2024-2025**

**Approved: Josephine T. Eduardo**

**Chair, CSD**

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**Dean, CICS**